

4.6 Summary and Notes

4.6.1 Summary

Idea.

1. If $\mathbb{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$ then $\|\mathbb{Z}\|^2 \sim \chi^2(r, \gamma)$ with $\gamma = \|\boldsymbol{\mu}\|^2$, mean $r + \gamma$ and variance $2r + 4\gamma$. The law is a $\text{Poisson}(\gamma/2)$ mixture of $\chi^2(r + 2k)$ laws, and its tails increase strictly with γ (Theorem 4.1.2 and Proposition 4.1.3).
2. $F(r, s, \gamma)$ and $t(s, \delta)$ put a noncentral numerator over an independent central denominator. The power of an F test increases strictly with γ , and $T^2 \sim F(1, s, \delta^2)$ (Theorem 4.1.4).
3. $\mathbf{E}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$ for any distribution. Under normality

$$\text{Var}(\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = 2 \text{tr}\{(\mathbf{A}\boldsymbol{\Sigma})^2\} + 4\boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}, \quad \text{Cov}(\mathbf{K}\mathbf{Y}, \mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}) = 2\mathbf{K}\boldsymbol{\Sigma}\mathbf{A}\boldsymbol{\mu}$$

(Theorem 4.2.3).

4. A normal quadratic form is a constant plus independent terms $\lambda_j(W_j + \nu_j)^2$, and for singular $\boldsymbol{\Sigma}$ possibly normal terms. The weights λ_j are the nonzero eigenvalues of $\mathbf{A}\boldsymbol{\Sigma}$ (Theorem 4.2.1).
5. For $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I})$, $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ is chi-squared iff $\mathbf{A}$ is idempotent. Then the degrees of freedom are $\text{rank } \mathbf{A} = \text{tr } \mathbf{A}$ and the noncentrality is $\boldsymbol{\mu}\mathbf{T}\mathbf{A}\boldsymbol{\mu}$ (Theorem 4.3.1). For positive definite $\boldsymbol{\Sigma}$ the condition is that $\mathbf{A}\boldsymbol{\Sigma}$ be idempotent. For projections, $\|\mathbf{P}\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(\text{rank } \mathbf{P}, \|\mathbf{P}\boldsymbol{\mu}\|^2/\sigma^2)$ (Theorem 4.3.2).
6. $\mathbf{B}\mathbf{Y}$ and $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ are independent if $\mathbf{B}\boldsymbol{\Sigma}\mathbf{A} = \mathbf{0}$, and $\mathbf{Y}\mathbf{T}\mathbf{A}\mathbf{Y}$ and $\mathbf{Y}\mathbf{T}\mathbf{B}\mathbf{Y}$ are independent if $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B} = \mathbf{0}$, because a quadratic form is a function of $\mathbf{A}\mathbf{Y}$ (Theorem 4.4.1 and Theorem 4.4.2).
7. Projections of a spherical normal vector onto mutually orthogonal subspaces are independent. This gives the independence of $\bar{Y}$ and S^2 and the exact $F(r - r_0, n - r, \gamma)$ law of the nested-model F statistic (Theorem 4.4.4, Corollary 4.4.5 and Theorem 4.4.6).
8. Cochran: if symmetric matrices add to an idempotent matrix and their ranks add up, the corresponding sums of squares are independent chi-squared variables (Theorem 4.5.1 and Theorem 4.5.2).

4.6.2 Notes and sources

Source texts. Rencher and Schaalje (2008, chapter 5) cover the material of [Section 4.2](#), [Section 4.3](#), [Section 4.4](#) and [Section 4.5](#) for positive definite covariance matrices. They derive the mgf of a quadratic form by integrating against the normal density and obtain its variance by differentiating the logarithm of the mgf; the converse of the chi-squared theorem is referred to the literature. They use the noncentrality $\boldsymbol{\mu}\mathbf{T}\boldsymbol{\mu}/2$. Seber and Lee (2003, section 2.4) reduce the general case to a spherical normal vector by an orthogonal diagonalization. They prove necessity of idempotence for central forms by factorizing the moment generating function, and give the eigenvalue criterion for singular covariance matrices. Christensen (2020, section 1.3) states the three conditions of Theorem 4.3.2(c) for arbitrary nonnegative definite $\boldsymbol{\Sigma}$, and obtains

independence by viewing quadratic forms as functions of linear ones, the approach followed in [Section 4.4](#). The canonical representation (Theorem 4.2.1), the proof of necessity through the coefficients Eq. 4.3.1, the monotonicity proof through the Poisson mixture, and the treatment of the Cochran algebra are this book's own arrangements of standard material.

Noncentral distributions. The noncentral chi-squared distribution first appeared in Fisher (1928). Patnaik (1949) studied its approximation by a scaled central chi-squared variable, and Satterthwaite (1946) used the same two-moment idea for linear combinations of mean squares, which reappears in Chapter 32. Johnson et al. (1995, chapters 29–31) collect the properties of the noncentral chi-squared, F and t families. The monotonicity of power in the noncentrality and in the degrees of freedom is treated systematically by Ghosh (1973).

General quadratic forms. The exact distribution of a quadratic form that is not chi-squared can be computed by numerical inversion of its characteristic function (Gil-Pelaez 1951; Imhof 1961). Mathai and Provost (1992) give a book-length treatment of quadratic forms in normal and non-normal variables. The mean square successive difference of Example (Two estimators of σ^2 under a drifting mean) goes back to von Neumann (1941), who derived the exact distribution of its ratio to S^2 .

Independence and Cochran's theorem. Cochran (1934) proved the decomposition theorem that bears his name, in the context of the analysis of covariance. The necessity of $\mathbf{A}\mathbf{\Sigma}\mathbf{B} = \mathbf{0}$ for independence of two forms is attributed to Craig (1943). Its full proof for arbitrary symmetric matrices and arbitrary means turned out to be subtle, and Driscoll and Gundberg (1986) trace the flawed and correct arguments. Searle (1971) gives general proofs of the independence and Cochran theorems in the style most linear-models texts follow. For nonnegative definite $\mathbf{A}$ and $\mathbf{B}$, the condition $\mathbf{\Sigma}\mathbf{A}\mathbf{\Sigma}\mathbf{B}\mathbf{\Sigma} = \mathbf{0}$ of Christensen (2020, Theorem 1.3.8) is in fact equivalent to $\mathbf{A}\mathbf{\Sigma}\mathbf{B} = \mathbf{0}$ (Exercise (B2)), so that theorem adds nothing to Theorem 4.4.2 even for singular $\mathbf{\Sigma}$, whereas his Theorem 1.3.9, for general symmetric $\mathbf{A}$ and $\mathbf{B}$, does. That the independence of $\bar{Y}$ and S^2 characterizes the normal distribution was shown by Lukacs (1942).

4.6.3 References

- Christensen, Ronald (2020). *Plane Answers to Complex Questions: The Theory of Linear Models*. 5th edition. Cham: Springer.
- Cochran, William G. (1934). The Distribution of Quadratic Forms in a Normal System, with Applications to the Analysis of Covariance. *Mathematical Proceedings of the Cambridge Philosophical Society* 30(2), 178–191.
- Craig, Allen T. (1943). Note on the Independence of Certain Quadratic Forms. *The Annals of Mathematical Statistics* 14(2), 195–197.
- Driscoll, Michael F. and Gundberg, William R. (1986). A History of the Development of Craig's Theorem. *The American Statistician* 40(1), 65–70.
- Fisher, Ronald A. (1928). The General Sampling Distribution of the Multiple Correlation Coefficient. *Proceedings of the Royal Society of London. Series A* 121(788), 654–673.
- Ghosh, B. K. (1973). Some Monotonicity Theorems for χ^2 , F and t Distributions with Applications. *Journal of the Royal Statistical Society. Series B* 35(3), 480–492.
- Gil-Pelaez, J. (1951). Note on the Inversion Theorem. *Biometrika* 38(3/4), 481–482.
- Imhof, J. P. (1961). Computing the Distribution of Quadratic Forms in Normal Variables. *Biometrika* 48(3/4), 419–426.
- Johnson, Norman L., Kotz, Samuel and Balakrishnan, N. (1995). *Continuous Univariate Distributions*. 2nd edition. New York: Wiley.
- Lukacs, Eugene (1942). A Characterization of the Normal Distribution. *The Annals of*

Mathematical Statistics 13(1), 91–93.

- Mathai, A. M. and Provost, Serge B. (1992). *Quadratic Forms in Random Variables: Theory and Applications*. New York: Marcel Dekker.
- Patnaik, P. B. (1949). The Non-Central χ^2 - and F-Distributions and Their Applications. *Biometrika* 36(1/2), 202–232.
- Rencher, Alvin C. and Schaalje, G. Bruce (2008). *Linear Models in Statistics*. 2nd edition. Hoboken, NJ: Wiley.
- Satterthwaite, Franklin E. (1946). An Approximate Distribution of Estimates of Variance Components. *Biometrics Bulletin* 2(6), 110–114.
- Searle, Shayle R. (1971). *Linear Models*. New York: Wiley.
- Seber, George A. F. and Lee, Alan J. (2003). *Linear Regression Analysis*. 2nd edition. Hoboken, NJ: Wiley.
- von Neumann, John (1941). Distribution of the Ratio of the Mean Square Successive Difference to the Variance. *The Annals of Mathematical Statistics* 12(4), 367–395.